

KOHLER SpatialAI

AI Prompt Architecture & Reliability Documentation

AI Bathroom Designer & Planner

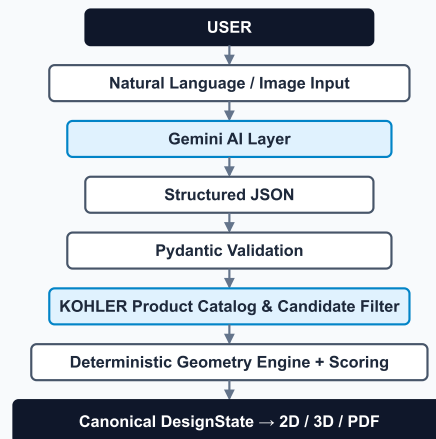
1. PURPOSE

KOHLER SpatialAI combines Generative AI with deterministic catalog, geometry, scoring, and optimization systems to create bathroom designs using real KOHLER catalog products. The AI is intentionally restricted to language-level intelligence. Product facts, spatial feasibility, scoring, placement, and canonical design state remain strictly controlled by deterministic application logic.

CORE OPERATING PRINCIPLE

AI proposes → Catalog verifies → Geometry validates → Scoring evaluates → Optimization decides → User remains in control

AI ARCHITECTURE



What AI Does

- Understand natural-language bathroom requirements
- Extract budget and category constraints
- Interpret style preferences
- Understand improvement & what-if requests
- Interpret supported room / image information
- Generate design explanations & trade-offs

What AI Does NOT Do

- Invent SKUs, prices, or dimensions
- Select nonexistent products
- Determine final physical coordinates
- Declare spatial feasibility or geometry
- Override catalog facts or scoring
- Directly mutate canonical DesignState

RELIABILITY PRINCIPLE: The LLM is treated as an untrusted language interface, not as the source of physical or commercial truth.

2. RUNTIME PROMPTS & AI RELIABILITY

The runtime LLM operates as a language-intelligence layer whose outputs must pass structured validation before reaching deterministic application services.

CORE SYSTEM PROMPT

You are the language-intelligence component of KOHLER SpatialAI. Your responsibility is to understand user intent and produce structured information that deterministic application services can safely consume. You may interpret natural-language requirements, identify budget and style preferences, extract category constraints, interpret improvement requests, and explain validated designs. You must not invent SKUs, prices, dimensions, availability, compatibility evidence, physical coordinates, or feasibility results. You must not override geometry, scoring, catalog data, or canonical DesignState. When information is unavailable or ambiguous, preserve uncertainty using null/unknown rather than guessing. Return only the requested structured format.

PROMPT ENGINEERING PRINCIPLES

PRINCIPLE	IMPLEMENTATION STRATEGY
Grounding	Product facts originate strictly from the KOHLER catalog dataset.
Least Authority	LLM receives only language-level responsibilities.
Structured Output	Responses are strictly validated through Pydantic schemas.
No Guessing	Missing or ambiguous values remain null/unknown.
Deterministic Finalization	Geometry, scoring, and physical placement occur completely outside the LLM.
User Control	AI suggestions become draft proposals, never silent mutations.

2.1 GEMINI KEY FAILOVER MECHANISM

KOHLER SpatialAI includes an ordered multi-key fallback mechanism for Gemini requests to prevent application downtime during API key exhaustion.



DETERMINISTIC DEGRADATION & RELIABILITY RULE

When all configured Gemini keys fail, supported budget expressions and category constraints are extracted deterministically. The application reports the degraded state without fabricating results.

KEY RULE: *AI failure must degrade functionality – never degrade truth.*

3. REQUIREMENT PARSING & CATALOG GROUNDING

The requirement parser converts natural-language user requests into structured constraints.

USER REQUEST

"I want a modern bathroom with a toilet below ₹20,000."

STRUCTURED INTERPRETATION

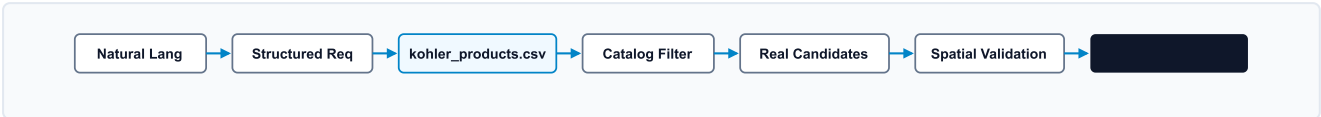
```
{ "style": "modern",  
  "category_constraints": { "toilet": {  
    "max_price_inr": 20000 } } }
```

The AI identifies the constraint but does not select the specific product SKU.

REQUIREMENT PARSER RULES

- Extract only information supported by the user's input message.
- Never invent missing room dimensions.
- Normalize dimensions to millimetres and monetary values to INR when unambiguous.
- Keep category constraints separate from global budget limits.
- Do not select products, generate final coordinates, or determine physical feasibility.

3.1 CATALOG GROUNDING PIPELINE



HARD GROUNDING RULE
If a product fact does not exist in the catalog, the AI cannot manufacture it.

3.2 PRODUCT RELATIONSHIPS & COMPATIBILITY

Product selection is not treated as isolated SKU matching. Inter-product relationships (e.g., Basin + Vanity forming a Wash Area, Faucet matching Basin, Shower Hardware wet-zone alignment) are validated outside the LLM using deterministic compatibility logic.

4. DETERMINISTIC GEOMETRY, SCORING & OPTIMIZATION

The geometry engine converts product dimensions and room constraints into validated physical layouts.

4.1 HARD-CONSTRAINT SPATIAL ENVELOPE VALIDATION

A vanity is not validated as an isolated rectangle; its WashStation installation envelope is checked against room boundaries, door swing, required clearances, fixture collisions, and circulation space.



CANONICAL STATE PRINCIPLE

One DesignState → One Geometry Definition → Multiple Renderers (2D Floorplan, 3D Twin, PDF Report).

Wall-mounted fixtures derive orientation from a canonical wall-anchor → rotation mapping, preventing 2D, 3D, and PDF renderers from independently inventing positions.

4.2 MULTI-OBJECTIVE SCORING ENGINE

METRIC	SMART BUDGET PROFILE	BALANCED PROFILE	PREMIUM PROFILE
Spatial Fit	30%	30%	30%
Budget Fit	40%	25%	15%
Style Match	15%	25%	30%
Compatibility	15%	20%	25%
Total Weight	100%	100%	100%

$$\text{Overall Score} = \frac{\sum(\text{Active Metric Weight} \times \text{Metric Score})}{\sum(\text{Active Metric Weights})}$$

* Unavailable metrics are excluded from both numerator and denominator rather than artificially set to zero. Accessibility & Sustainability are secondary informational indicators.

5. AI OPTIMIZER, WHAT-IF & ENGINEERING PROOF

The Copilot interprets user improvement requests and generates safe draft proposals.

5.1 AI OPTIMIZER & WHAT-IF EXECUTION FLOW

COPILOT PROPOSAL WORKFLOW

Current Design ↓ User Request ("Lower Cost") ↓ AI Intent Interpretation ↓ Catalog Search & Spatial Validation ↓ Draft Proposal Generated ↓ User Decision (ACCEPT → Update / DISCARD → Retain)

WHAT-IF EXECUTION EXAMPLE

User: "Move shower to opposite wall."
AI Output: { "intent": "wall_change", "category": "shower", "placement_preference": "other_feasible_wall" }
Deterministic Engine: Position Generation → Room Boundary Check → Collision Validation → Wet-Zone Check → Clearance → Score Recalculation.

5.2 EXPLAINABILITY PROMPT SPECIFICATION

Explain the current validated bathroom design using ONLY supplied DesignState, catalog facts, validation results, and scores. Explain: selected products, requirement satisfaction, spatial evidence, budget implications, style matching, and trade-offs. Never invent specifications, prices, dimensions, geometry, or scores. Do not independently recalculate score values.

5.3 AUTOMATED TESTING & ENGINEERING PROOF

The engineering suite includes automated tests covering all critical deterministic application boundaries.

Validated Engine Rules

- Fixture overlap & clearance limits
- Door swing collision avoidance
- Rotation & orientation logic
- Budget limit enforcement

Multi-Room Robustness

- Missing metric handling
- Canonical DesignState consistency
- Multi-objective scoring weights
- Spatial feasibility verification

CORE RELIABILITY STORY

The LLM can misunderstand language, but it cannot silently rewrite physical truth.

6. SAFETY, PROMPT INVENTORY & EVALUATOR SUMMARY

6.1 COMPLETE SYSTEM RESPONSIBILITY MATRIX

SYSTEM LAYER	PRIMARY ARCHITECTURAL RESPONSIBILITY
Gemini AI	Natural language interpretation, constraint extraction, conversational trade-off explanations.
Fallback Parser	Deterministic constraint extraction when external AI service fails or is unreachable.
KOHLER Catalog	Commercial ground truth: SKUs, pricing, dimensions, finishes, collections.
Geometry Engine	Physical spatial feasibility: clearances, boundary validation, wet zones, collision checks.
Compatibility Logic	Inter-product structural and plumbing compatibility validation.
Scoring Engine	Multi-objective quantitative design score evaluation.
Optimizer	Candidate selection and algorithmic spatial placement optimization.
DesignState	Canonical single-source-of-truth JSON structure representing the complete bathroom.
2D / 3D / PDF	Deterministic visualization rendering driven strictly by canonical DesignState.
User	Final design selection, review, acceptance, and approval control.

6.2 MASTER PROMPT INVENTORY

Runtime System Instruction: Defines LLM authority limits	Requirement Parser: Converts NL to JSON constraints
Price Constraint Logic: Extracts category/global budget limits	Style Interpretation: Converts style keywords into design criteria
AI Optimizer: Generates validated improvement proposals	Explainability Prompt: Generates evidence-based design rationale
What-If Interpreter: Converts conversational changes to intent	Fallback Parser: Maintains deterministic degradation on AI fail

6.3 KEY EVALUATOR QUESTIONS

Why use an LLM? To interpret nuanced NL requirements and explain design trade-offs.	Why not let LLM handle geometry? Physical feasibility requires exact, deterministic geometric validation.
Can AI invent products/prices? No. Product selection and facts are restricted to the physical CSV catalog.	What happens if Gemini fails? Ordered API-key failover and deterministic fallback parsing prevent failure.

FINAL ARCHITECTURE PIPELINE

AI PROPOSES → CATALOG VERIFIES → GEOMETRY VALIDATES → SCORING EVALUATES → OPTIMIZATION DECIDES → USER CONTROLS

Final Statement: KOHLER SpatialAI applies Generative AI where language intelligence provides the most value, while reserving product truth, geometric feasibility, scoring, optimization, and canonical state management for deterministic and testable application logic. The result is a constraint-aware AI bathroom planning system designed to produce explainable, catalog-grounded, physically validated designs.